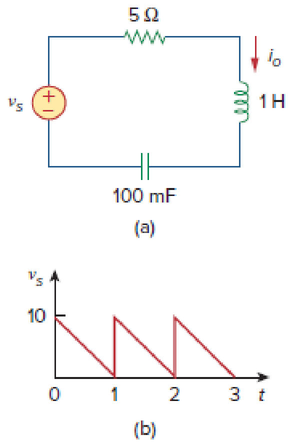
Chapter 17, Problem 36P

Problem

Find the response *io* for the circuit in Fig. 17.72(a), where *vs*(*t*) is shown in Fig. 17.72(b).

Figure 17.72:



Step-by-step solution

Step 1 of 6

Refer to Figure 17.36 in the textbook for the circuit and waveform.

From the waveform, the time period is, $T = 1\text{ s}$.

Therefore, the angular frequency is,

$$\begin{aligned}\omega_0 &= \frac{2\pi}{T} \\ &= \frac{2\pi}{1} \\ &= 2\pi \text{ rad/s}\end{aligned}$$

Express the periodic voltage waveform in Fourier series. Write the general expression of the Fourier series.

$$v_s(t) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t\right) \quad \text{......} \; (1)$$

Here, $a_0, a_n, \text{and } b_n$ are coefficients of Fourier series.

Calculate the value of the Fourier coefficient a_0 of Fourier series.

$$\begin{aligned}a_0 &= \frac{1}{T}\int_0^T v_s(t)\,dt \\ &= \frac{1}{1}\int_0^1 10(1-t)\,dt \\ &= 10\left[t - \frac{t^2}{2}\right]_0^1 \\ &= 10\left[1 - \frac{1}{2}\right] \\ &= 5\end{aligned}$$

Step 2 of 6

Calculate the value of the Fourier coefficient a_n of Fourier series.

$$a_n = \frac{2}{T}\int_0^T v_s(t)\cos n\omega_0 t\,dt$$
$$= \frac{2}{1}\int_0^1 10(1-t)\cos n(2\pi)t\,dt$$
$$= 20\int_0^1 (1-t)\cos 2\pi nt\,dt$$
$$= 20\left[\frac{1}{2\pi n}\sin 2n\pi t - \frac{1}{4\pi^2 n^2}\cos 2n\pi t - \frac{t}{2n\pi}\sin 2n\pi t\right]_0^1$$
$$= 20\left[-\frac{1}{4\pi^2 n^2} + -\frac{1}{4\pi^2 n^2}\right]$$
$$= 0$$

Step 3 of 6

Calculate the value of the Fourier coefficient b_n of Fourier series.

$$b_n = \frac{2}{T}\int_0^T v_s(t)\sin n\omega_0 t\,dt$$
$$= \frac{2}{1}\int_0^1 10(1-t)\sin n\omega_0 t\,dt$$
$$= 20\int_0^1 (1-t)\sin n\omega_0 t\,dt$$
$$= 20\left[-\frac{1}{2n\pi}\cos(2n\pi t) - \frac{1}{4n^2\pi^2}\sin(2n\pi t) + \frac{t}{2n\pi}\cos(2n\pi t)\right]_0^1$$
$$= 20\left[-\frac{1}{2n\pi} - 0 + \frac{1}{2n\pi} - \left(-\frac{1}{2n\pi} - 0 + 0\right)\right]$$
$$= 20\left[\frac{1}{2n\pi}\right]$$
$$= \frac{10}{n\pi}$$

Step 4 of 6

Recall equation (1).

$$v_s(t) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos n\omega_0 t + b_n \sin n\omega_0 t\right)$$

Substitute the values of the coefficients a_0, a_n and b_n in equation (1).

$$\begin{aligned} v_s(t) &= 5 + \sum_{n=1}^{\infty}\left((0)\cos n2\pi t + \frac{10}{n\pi}\sin n2\pi t\right) \\ &= 5 + \sum_{n=1}^{\infty}\left(\frac{10}{n\pi}\sin 2n\pi t\right)\text{ V} \end{aligned}$$

Calculate the impedance of the 1H inductor.

$$\begin{aligned} Z_L &= j\omega_n L \\ &= j\omega_n(1) \\ &= j\omega_n \end{aligned}$$

Calculate the impedance of the 100 mF capacitor.

$$\begin{aligned} Z_C &= \frac{1}{j\omega_n C} \\ &= \frac{1}{j\omega_n\left(100\times10^{-3}\right)} \\ &= -j\frac{10}{\omega_n} \end{aligned}$$

Step 5 of 6

From the Figure 17.72(a), the current I_o is,

$$\begin{aligned} I_o &= \frac{V_S}{5 + Z_L + Z_C} \\ &= \frac{V_S}{5 + j\omega_n - \frac{j10}{\omega_n}} \end{aligned}$$

For dc component, $\omega_n = 0$.

Therefore,

$$\begin{aligned} I_0 &= \frac{V_S}{5 + j(0) - \frac{j10}{0}} \\ &= 0 \end{aligned}$$

Step 6 of 6

For the nth harmonic,

$$I_o = \frac{\frac{10}{n\pi}\angle 0^\circ}{5 + j2\pi n - \frac{j10}{2\pi n}}$$

$$= \frac{10}{5n\pi + j\left(2n^2\pi^2 - 5\right)}$$

$$= \frac{10}{\sqrt{25n^2\pi^2 + \left(2n^2\pi^2 - 5\right)^2}}\angle -\tan^{-1}\left(\frac{2n^2\pi^2 - 5}{25n^2\pi^2}\right)$$

$$= A_n \angle -\phi_n$$

The response of the circuit is,

$$i_o(t) = \sum_{n=1}^{\infty} A_n \sin\left(2n\pi t + \phi_n\right)$$

Here,

$$A_n = \frac{10}{\sqrt{25n^2\pi^2 + \left(2n^2\pi^2 - 5\right)^2}}$$

$$\phi_n = -\tan^{-1}\left(\frac{2n^2\pi^2 - 5}{25n^2\pi^2}\right)$$

Therefore, the response of the circuit, $i_o(t)$ is $\boxed{\sum_{n=1}^{\infty} A_n \sin\left(2n\pi t + \phi_n\right)}$.